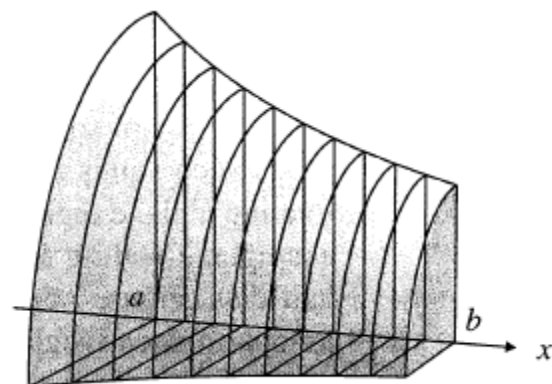


Volumes by Slicing



To be specific, suppose that the solid S lies between planes perpendicular to the x -axis at positions $x = a$ and $x = b$ and that the cross-sectional area of S in the plane perpendicular to the x -axis at x is a known function $A(x)$, for $a \leq x \leq b$. We assume that $A(x)$ is continuous on $[a, b]$. If $a = x_0 < x_1 < x_2 < \cdots < x_{n-1} < x_n = b$, then $P = \{x_0, x_1, x_2, \dots, x_{n-1}, x_n\}$ is a partition of $[a, b]$ into n subintervals, and the planes perpendicular to the x -axis at $x_1, x_2, \dots, x_{n-1}$ divide the solid into n slices

of which the i th has thickness $\Delta x_i = x_i - x_{i-1}$. The volume ΔV_i of that slice lies between the maximum and minimum values of $A(x) \Delta x_i$ for values of x in $[x_{i-1}, x_i]$, so

$$\Delta V_i = A(c_i) \Delta x_i$$

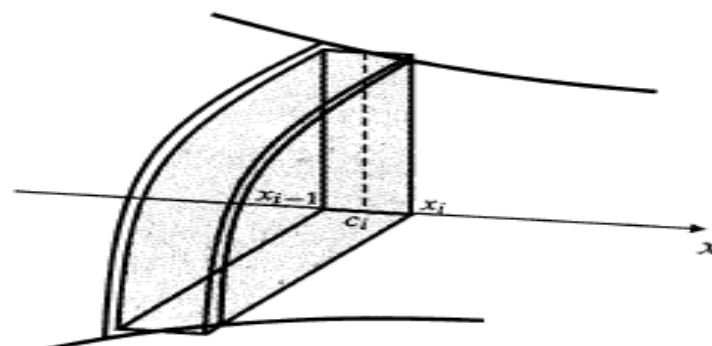
for some c_i in $[x_{i-1}, x_i]$, by the Intermediate-Value Theorem. The volume of the solid is therefore given by the Riemann sum

$$V = \sum_{i=1}^n \Delta V_i = \sum_{i=1}^n A(c_i) \Delta x_i.$$

Letting n approach infinity in such a way that $\max \Delta x_i$ approaches 0, we obtain the definite integral of $A(x)$ over $[a, b]$ as the limit of this Riemann sum. Therefore:

The volume V of a solid between $x = a$ and $x = b$ having cross-sectional area $A(x)$ at position x is

$$V = \int_a^b A(x) dx.$$

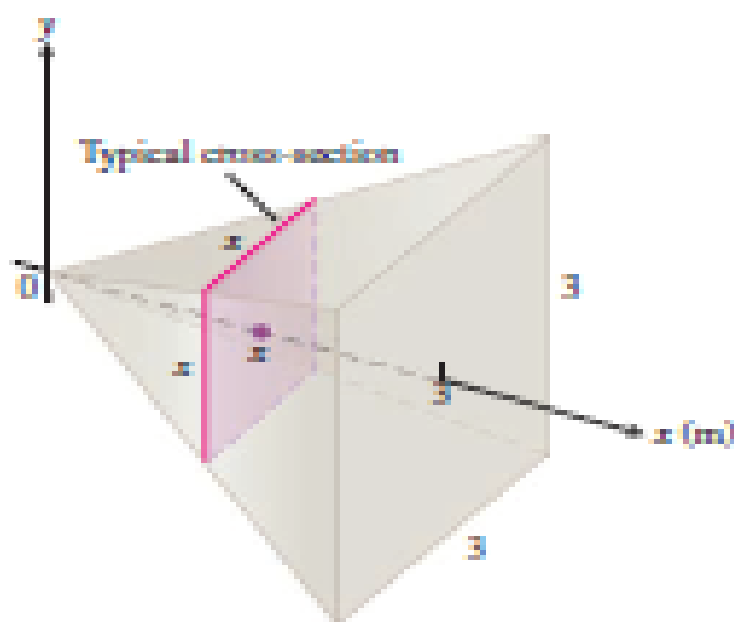


The volume of a slice

EXAMPLE 1 A pyramid 3 m high has a square base that is 3 m on a side. The cross-section of the pyramid perpendicular to the altitude x m down from the vertex is a square x m on a side. Find the volume of the pyramid.

Solution

1. *A sketch.* We draw the pyramid with its altitude along the x -axis and its vertex at the origin and include a typical cross-section



2. *A formula for $A(x)$.* The cross-section at x is a square x meters on a side, so its area is

$$A(x) = x^2.$$

3. *The limits of integration.* The squares lie on the planes from $x = 0$ to $x = 3$.

4. *Integrate to find the volume:*

$$V = \int_0^3 A(x) \, dx = \int_0^3 x^2 \, dx = \left. \frac{x^3}{3} \right|_0^3 = 9 \text{ m}^3.$$

Solids of Revolution

If the region R bounded by $y = f(x)$, $y = 0$, $x = a$, and $x = b$ is rotated about the x -axis, then the cross-section of the solid generated in the plane perpendicular to the x -axis at x is a circular disk of radius $|f(x)|$. The area of this cross-section is $A(x) = \pi(f(x))^2$, so the volume of the solid of revolution is

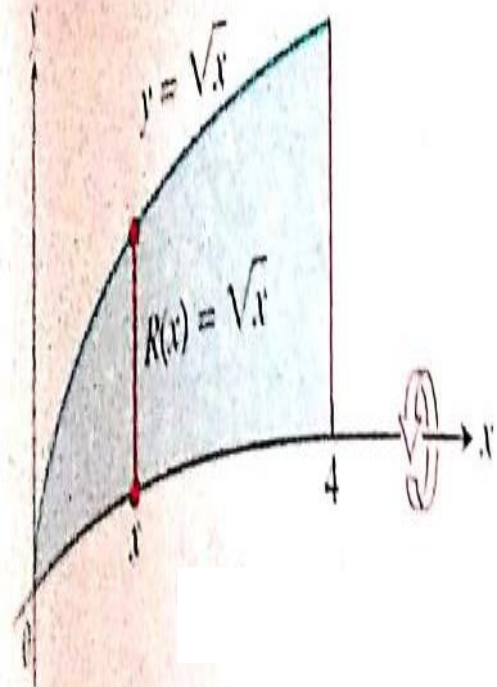
$$V = \pi \int_a^b (f(x))^2 dx.$$

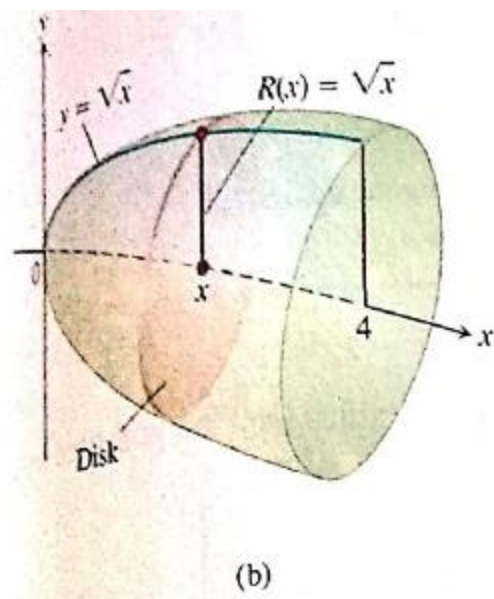
Solids of Revolution: The Disk Method

The solid generated by rotating (or revolving) a plane region about an axis in its plane is called a **solid of revolution**. To find the volume of a solid we need only observe that the cross-sectional area $A(x)$ is the area of a disk of radius $R(x)$, the distance of the planar region's boundary from the axis of revolution. The area is then

$$A(x) = \pi(\text{radius})^2 = \pi[R(x)]^2.$$

So the definition of volume in this case gives





Volume by Disks for Rotation About the x -axis

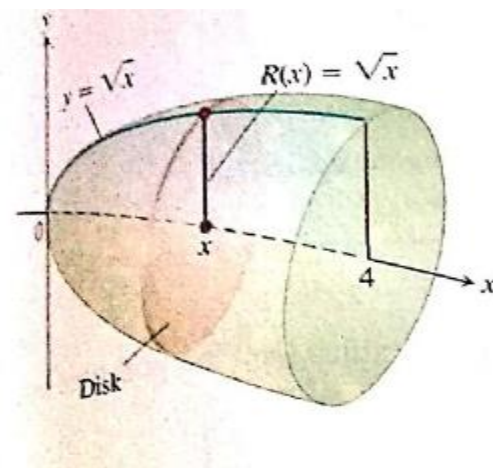
$$V = \int_a^b A(x) \, dx = \int_a^b \pi [R(x)]^2 \, dx.$$

EXAMPLE The region between the curve $y = \sqrt{x}$, $0 \leq x \leq 4$, and the x -axis is revolved about the x -axis to generate a solid. Find its volume.

Solution We draw figures showing the region, a typical radius, and the generated solid.
The volume is

$$\begin{aligned} V &= \int_a^b \pi [R(x)]^2 dx \\ &= \int_0^4 \pi [\sqrt{x}]^2 dx \\ &= \pi \int_0^4 x dx = \pi \left[\frac{x^2}{2} \right]_0^4 = \pi \frac{(4)^2}{2} = 8\pi. \end{aligned}$$

Radius $R(x) = \sqrt{x}$ for
rotation around x -axis



EXAMPLE The circle

$$x^2 + y^2 = a^2$$

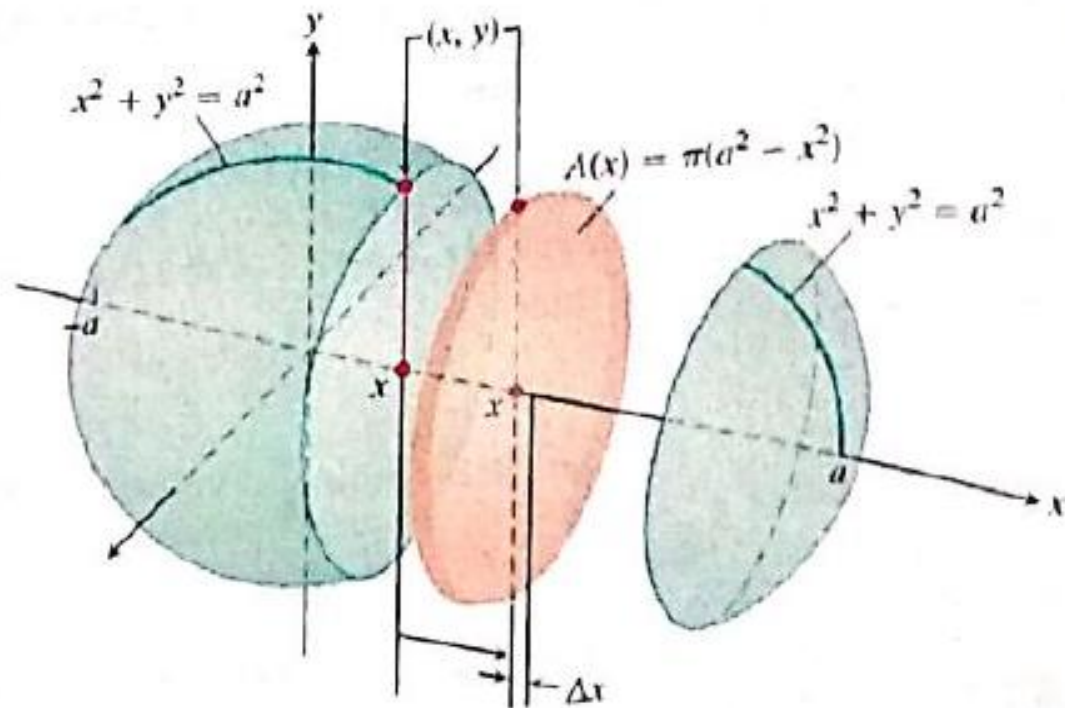
is rotated about the x -axis to generate a sphere. Find its volume.

Solution We imagine the sphere cut into thin slices by planes perpendicular to the x -axis. The cross-sectional area at a typical point x between $-a$ and a is

$$A(x) = \pi y^2 = \pi(a^2 - x^2), \quad R(x) = \sqrt{a^2 - x^2} \text{ for rotation around } x\text{-axis}$$

Therefore, the volume is

$$V = \int_{-a}^a A(x) \, dx = \int_{-a}^a \pi(a^2 - x^2) \, dx = \pi \left[a^2 x - \frac{x^3}{3} \right]_{-a}^a = \frac{4}{3} \pi a^3.$$



The sphere generated by rotating the circle $x^2 + y^2 = a^2$ about the x-axis. The radius is $R(x) = y = \sqrt{a^2 - x^2}$.

EXAMPLE Find the volume of the solid generated by revolving the region bounded by $y = \sqrt{x}$ and the lines $y = 1, x = 4$ about the line $y = 1$.

Solution We draw figures showing the region, a typical radius, and the generated solid. The volume is

$$V = \int_1^4 \pi [R(x)]^2 dx$$

$$= \int_1^4 \pi [\sqrt{x} - 1]^2 dx$$

$$= \pi \int_1^4 [x - 2\sqrt{x} + 1] dx$$

$$= \pi \left[\frac{x^2}{2} - 2 \cdot \frac{2}{3} x^{3/2} + x \right]_1^4 = \frac{7\pi}{6}.$$

Radius $R(x) = \sqrt{x} - 1$
for rotation around $y = 1$

Expand integrand

Integrate.

To find the volume of a solid generated by revolving a region between the y -axis and a curve $x = R(y)$, $c \leq y \leq d$, about the y -axis, we use the same method with x replaced by y . In this case, the circular cross-section is

$$A(y) = \pi[\text{radius}]^2 = \pi[R(y)]^2,$$

and the definition of volume gives

Volume by Disks for Rotation About the y -axis

$$V = \int_c^d A(y) \, dy = \int_c^d \pi[R(y)]^2 \, dy.$$

EXAMPLE Find the volume of the solid generated by revolving the region between the y -axis and the curve $x = 2/y$, $1 \leq y \leq 4$, about the y -axis.

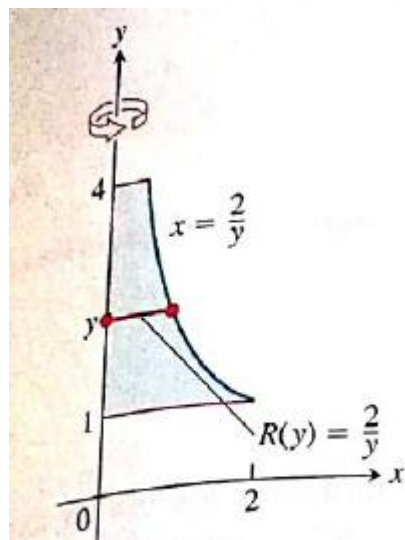
Solution We draw figures showing the region, a typical radius, and the generated solid. The volume is

$$V = \int_1^4 \pi [R(y)]^2 dy$$

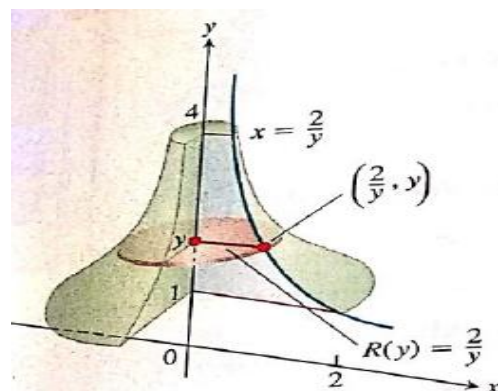
$$= \int_1^4 \pi \left(\frac{2}{y}\right)^2 dy$$

Radius $R(y) = \frac{2}{y}$ for
rotation around y -axis

$$= \pi \int_1^4 \frac{4}{y^2} dy = 4\pi \left[-\frac{1}{y}\right]_1^4 = 4\pi \left[\frac{3}{4}\right] = 3\pi.$$



(a)



(b)

EXAMPLE Find the volume of the solid generated by revolving the region between the parabola $x = y^2 + 1$ and the line $x = 3$ about the line $x = 3$.

Solution We draw figures showing the region, a typical radius, and the generated solid. Note that the cross-sections are perpendicular to the line $x = 3$ and have y -coordinates from $y = -\sqrt{2}$ to $y = \sqrt{2}$. The volume is

$$V = \int_{-\sqrt{2}}^{\sqrt{2}} \pi [R(y)]^2 dy$$

$$y = \pm \sqrt{2} \text{ when } x = 3$$

$$= \int_{-\sqrt{2}}^{\sqrt{2}} \pi [2 - y^2]^2 dy$$

$$\text{Radius } R(y) = 3 - (y^2 + 1) \\ \text{for rotation around axis } x = 3$$

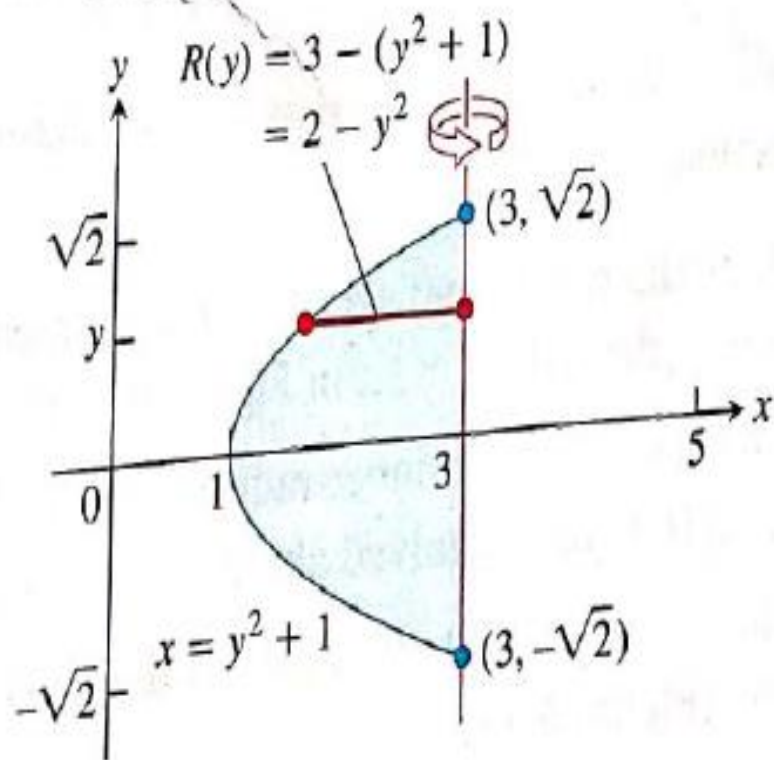
$$= \pi \int_{-\sqrt{2}}^{\sqrt{2}} [4 - 4y^2 + y^4] dy$$

Expand integrand.

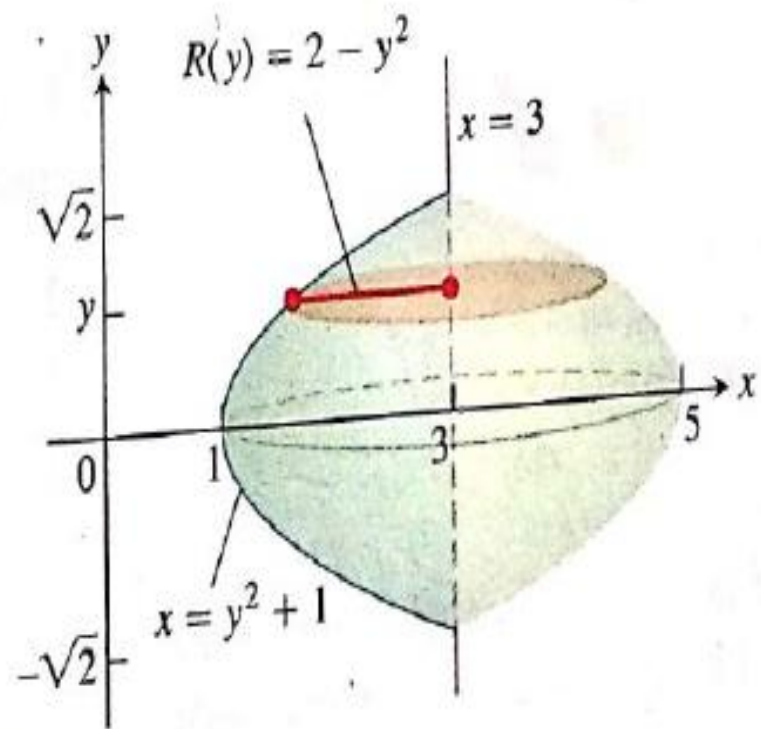
$$= \pi \left[4y - \frac{4}{3}y^3 + \frac{y^5}{5} \right]_{-\sqrt{2}}^{\sqrt{2}}$$

Integrate.

$$= \frac{64\pi\sqrt{2}}{15}.$$



(a)



(b)

Solids of Revolution: The Washer Method

If the region we revolve to generate a solid does not border on or cross the axis of revolution, the solid has a hole in it. The cross-sections perpendicular to the axis of revolution are *washers* (instead of disks). The dimensions of a typical washer are

Outer radius: $R(x)$

Inner radius: $r(x)$

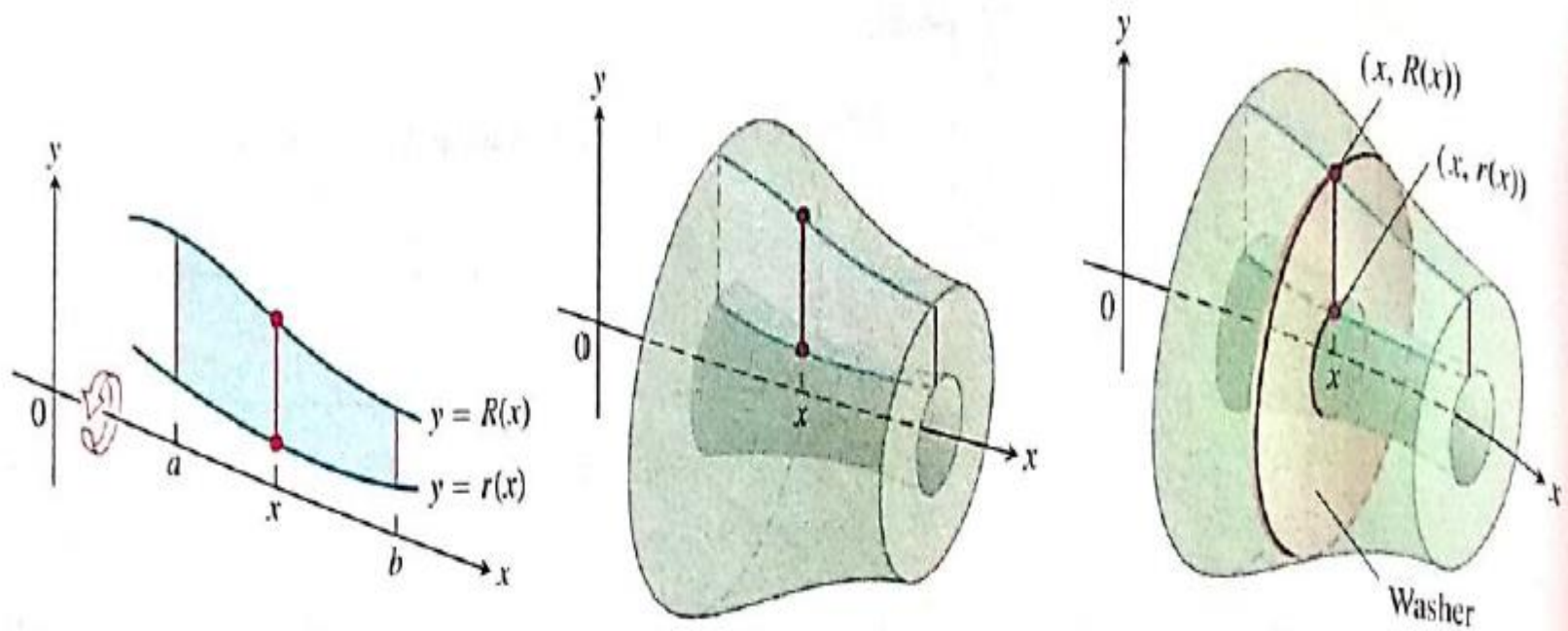
The washer's area is

$$A(x) = \pi[R(x)]^2 - \pi[r(x)]^2 = \pi([R(x)]^2 - [r(x)]^2).$$

Consequently, the definition of volume in this case gives

Volume by Washers for Rotation About the x-axis

$$V = \int_a^b A(x) dx = \int_a^b \pi([R(x)]^2 - [r(x)]^2) dx.$$



The cross-sections of the solid of revolution generated here are washers, not disks, so the integral $\int_a^b A(x) dx$ leads to a slightly different formula.

This method for calculating the volume of a solid of revolution is called the **washer method** because a thin slab of the solid resembles a circular washer of outer radius $R(x)$ and inner radius $r(x)$.

EXAMPLE The region bounded by the curve $y = x^2 + 1$ and the line $y = -x + 3$ is revolved about the x -axis to generate a solid. Find the volume of the solid.

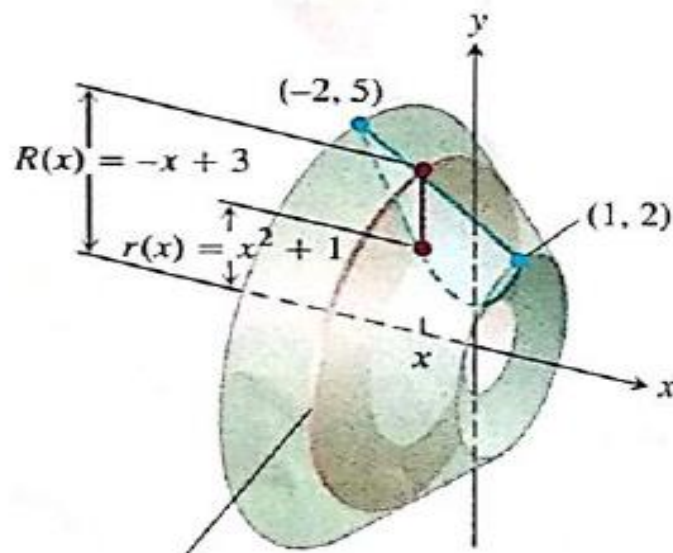
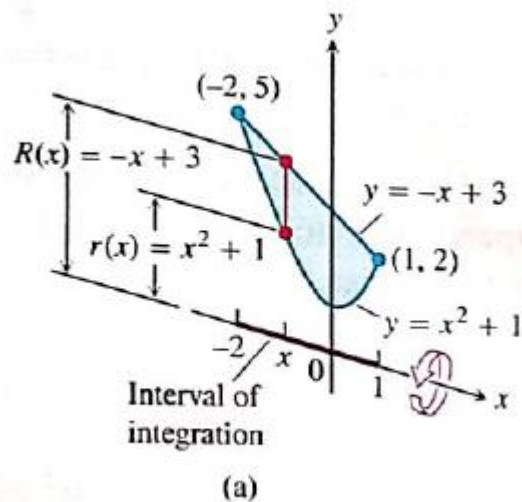
Solution We use the four steps for calculating the volume of a solid as discussed early in this section.

1. Draw the region and sketch a line segment across it perpendicular to the axis of revolution (the red segment in Figure
2. Find the outer and inner radii of the washer that would be swept out by the line segment if it were revolved about the x -axis along with the region.

These radii are the distances of the ends of the line segment from the axis of revolution

$$\text{Outer radius: } R(x) = -x + 3$$

$$\text{Inner radius: } r(x) = x^2 + 1$$



Washer cross-section

Outer radius: $R(x) = -x + 3$

Inner radius: $r(x) = x^2 + 1$

(b)

(a) The region in Example spanned by a line segment perpendicular to the axis of revolution. (b) When the region is revolved about the x -axis, the line segment generates a washer.

3. Find the limits of integration by finding the x -coordinates of the intersection points of the curve and line in Figure

$$x^2 + 1 = -x + 3$$

$$x^2 + x - 2 = 0$$

$$(x + 2)(x - 1) = 0$$

$$x = -2, \quad x = 1$$

Limits of integration

4. Evaluate the volume integral.

$$V = \int_a^b \pi([R(x)]^2 - [r(x)]^2) dx$$

Rotation around x -axis

$$= \int_{-2}^1 \pi((-x + 3)^2 - (x^2 + 1)^2) dx$$

Values from Steps 2
and 3

$$= \pi \int_{-2}^1 (8 - 6x - x^2 - x^4) dx$$

Simplify algebraically.

$$= \pi \left[8x - 3x^2 - \frac{x^3}{3} - \frac{x^5}{5} \right]_{-2}^1 = \frac{117\pi}{5}$$



EXAMPLE The region bounded by the parabola $y = x^2$ and the line $y = 2x$ in the first quadrant is revolved about the y -axis to generate a solid. Find the volume of the solid.

Solution First we sketch the region and draw a line segment across it perpendicular to the axis of revolution (the y -axis).

The radii of the washer swept out by the line segment are $R(y) = \sqrt{y}$, $r(y) = y/2$

The line and parabola intersect at $y = 0$ and $y = 4$, so the limits of integration are $c = 0$ and $d = 4$. We integrate to find the volume:

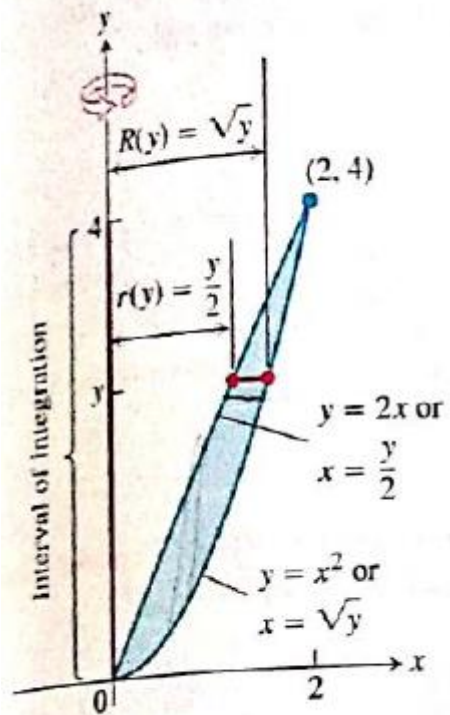
$$V = \int_c^d \pi ([R(y)]^2 - [r(y)]^2) dy$$

Rotation around y -axis

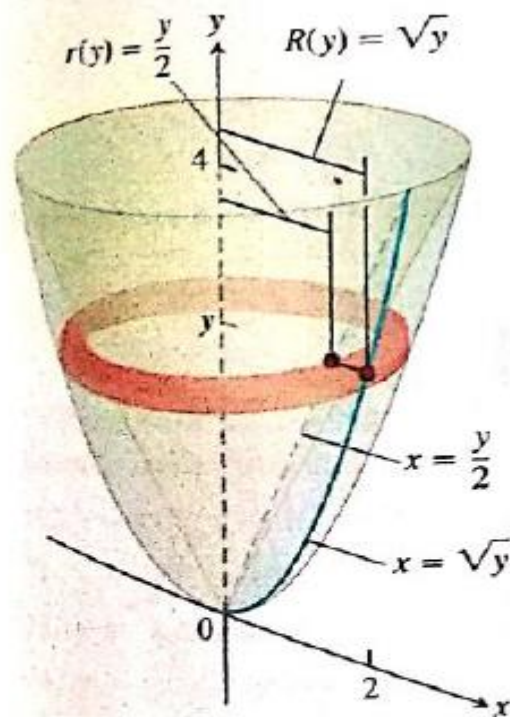
$$= \int_0^4 \pi \left([\sqrt{y}]^2 - \left[\frac{y}{2} \right]^2 \right) dy$$

Substitute for radii and limits of integration.

$$= \pi \int_0^4 \left(y - \frac{y^2}{4} \right) dy = \pi \left[\frac{y^2}{2} - \frac{y^3}{12} \right]_0^4 = \frac{8}{3} \pi.$$



(a)



(b)